FLAT ESSENTIALLY FINITE TYPE ALGEBRAS AND MODULES OVER A VALUATION RING ARE ESSENTIALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Writing an essentially finite type algebra as a quotient of a localisation of a polynomial algebra, and finite presentation over a subring. Algebra, Lemma \ref{algebra-lemma-finitely-presented-over-subring}, cited at proof lines 6023-6025. Set up the presentation and explain why finite presentation over the localisation is what has to be proved.
- The kernel of a presentation localises from its intersection with the polynomial algebra. Algebra, Lemma \ref{algebra-lemma-submodule-localization}, cited at proof lines 6029-6030. Identifies the kernel with the localisation of the constructed intersection.
- Torsion free modules over a valuation ring are flat, and a flat finite module of essentially finite type over a valuation ring is finitely presented. Lemma \ref{lemma-valuation-ring-torsion-free-flat} and Lemma \ref{lemma-flat-finite-type-valuation-ring-finite-presentation}, cited at proof lines 6033 and 6035-6036. Give flatness and then finite presentation of the quotient module.
- Localisation is exact. Algebra, Proposition \ref{algebra-proposition-localization-exact}, cited at proof lines 6037-6038. Concludes that the localisation of the constructed submodule is the module itself.

Source reference. Original lemma 0GSE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20929–20929. The lemma's own canonical tag,

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quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0561: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6694–6694. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00CU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 474–474. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0539: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6597–6597. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 053E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6602–6602. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00CS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 472–472. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a valuation ring. Let $A \to B$ be a local homomorphism which is essentially of finite type. Let M be a finite B-module.*

(1) *If B is flat over A, then B is essentially of finite presentation over A.*
(2) *If M is flat as an A-module, then M is finitely presented as a B-module.*

Proof. By assumption we can write B as a quotient of the localization of a polynomial algebra $P = A[x_1, \ldots, x_n]$ at a prime ideal $\mathfrak{q}$. In case (1) we consider $M = B$ as a finite module over $P_\mathfrak{q}$ and in case (2) we consider M as a finite module over $P_\mathfrak{q}$. In both cases, we have to show that this is a finitely presented $P_\mathfrak{q}$-module, see Algebra, Lemma 0561 for case (2).

Choose a presentation $0 \to K \to P_\mathfrak{q}^{\oplus r} \to M \to 0$ which is possible because M is finite over $P_\mathfrak{q}$. Let $L = P^{\oplus r} \cap K$. Then $K = L_\mathfrak{q}$, see Algebra, Lemma 00CU. Then $N = P^{\oplus r}/L$ is a submodule of M and hence flat by Lemma 0539. Since also N is a finite P-module, we see that N is finitely presented as a P-module by Lemma 053E. Since localization is exact (Algebra, Proposition 00CS) we see that $N_\mathfrak{q} = M$ and we conclude. □